Deploy Your Own Cryptocurrency

Hands-On ERC-20 Lab & Deep Dive

Standalone Technical Lab

"From zero to token in 15 minutes"

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University Lecture Series

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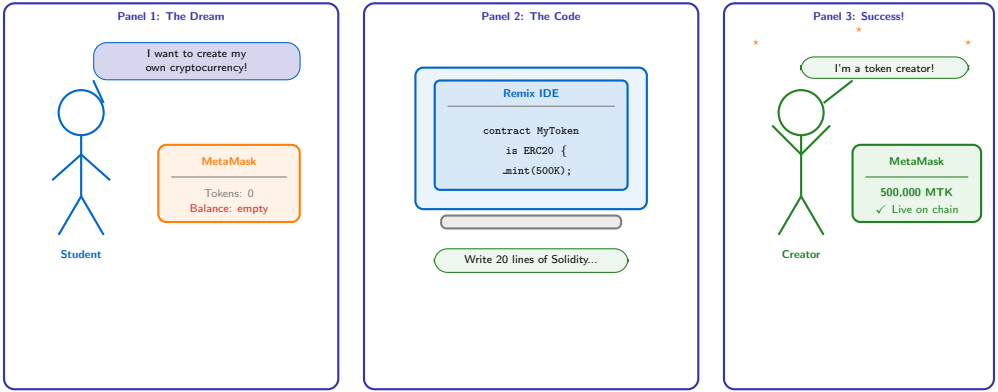
Part 1: Deploy Your Own Cryptocurrency

A step-by-step hands-on lab using Remix IDE and Sepolia testnet

The Mission



Creating a token is writing a smart contract — anyone can do it in minutes.

Step 1 — The Complete Contract

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
import "@openzeppelin/contracts/access/Ownable.sol";

contract MyToken is ERC20, Ownable {
    uint256 public constant MAX_SUPPLY
        = 1_000_000 * 10**18;

    constructor()
        ERC20("MyToken", "MTK")
        Ownable(msg.sender)
    {
        _mint(msg.sender, 500_000 * 10**18);
    }

    function mint(address to, uint256 amount)
        public onlyOwner
    {
        require(
            totalSupply() + amount <= MAX_SUPPLY,
            "Exceeds cap"
        );
        _mint(to, amount);
    }

    function burn(uint256 amount) public {
        _burn(msg.sender, amount);
    }
}
```

Name & Symbol
"MyToken", "MTK"
Human-readable identity

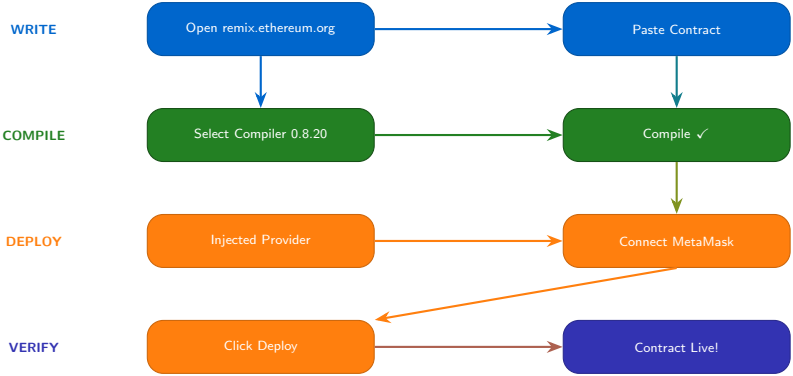
Initial Supply
500K of 1M cap (50%)
Minted to deployer

Mint (Owner Only)
Can mint remaining 500K
Enforces MAX_SUPPLY cap

Burn (Anyone)
Any holder can burn
their own tokens

This contract gives you a capped, mintable, burnable ERC-20 token — 500K minted now, 500K mintable later.

Step 2 — Deploy on Remix



Remix IDE handles compilation and deployment — no local tooling required.

Step 3 — Verify & Interact

Panel 1: Contract is Live

Etherscan

`0x7a3B...9F2d`

MyToken (MTK)

ERC-20 Token ✓ Verified

Your contract is live!

On-chain forever

Panel 2: Read Contract

balanceOf(yourAddress)

`Returns:`

500000000000000000000000

That's 500K tokens
with 18 decimals!

$500{,}000 \times 10^{18}$
= 500K human-readable

Panel 3: Write Contract

transfer()

`to: 0xFriend...`

`amount: 100*10^18`

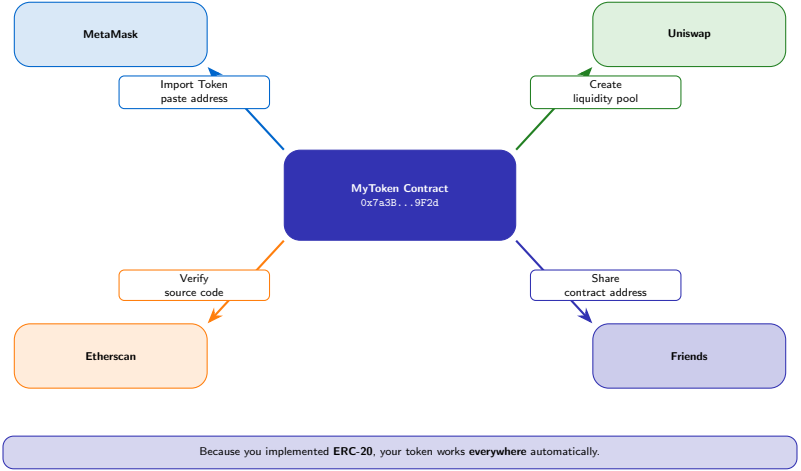
✓ **TX Success!**
Sent 100 MTK!

Gas used: ∼65,000

Confirmed in ∼12 sec

Once deployed, anyone can read your contract and interact with your token.

Step 4 — Add to MetaMask & Share



ERC-20 compliance means instant compatibility with the entire Ethereum ecosystem.

Your Token Deployment Recipe

Token Deployment Recipe

Ingredients:

- ✓ MetaMask wallet (with Sepolia test ETH)
- ✓ Web browser (Chrome or Firefox)
- ✓ Remix IDE (remix.ethereum.org)
- ✓ Contract code (from Step 1)

Steps:

1. Open `remix.ethereum.org` in your browser
2. Create a new file → name it `MyToken.sol`
3. Paste the complete contract code (from Frame 4)
4. Select Solidity compiler version `0.8.20`
5. Click "Compile MyToken.sol" → wait for ✓
6. Switch Environment to "Injected Provider – MetaMask"
7. Ensure MetaMask is on **Sepolia** testnet
8. Click "Deploy" → confirm transaction in MetaMask
9. Copy your contract address from the console
10. Go to `sepolia.etherscan.io` → verify it's live!

Yield: 1 fully functional ERC-20 token on Sepolia testnet

Prep time: 5 min • Deploy time: 2 min • Verification: 1 min • Total: under 15 minutes

Difficulty: Beginner • Cost: Free (testnet) • Serves: unlimited tokens

Follow this recipe step by step — you will have your own token deployed in under 15 minutes.

Part 2: Deep Dive — The Technical Details

Formal explanations behind five key concepts from the visual introduction

How Tokens Actually Work — The Balance Mapping

State: _balances

Address	Balance
0xAlice	500,000
0xBob	10,000
0xCarol	5,000

Total: 515,000 MTK

transfer(Bob, 100)

require(500K >= 100) ✓

Alice: 500,000

−100 = 499,900

Bob: 10,000

+100 = 10,100

After Transfer

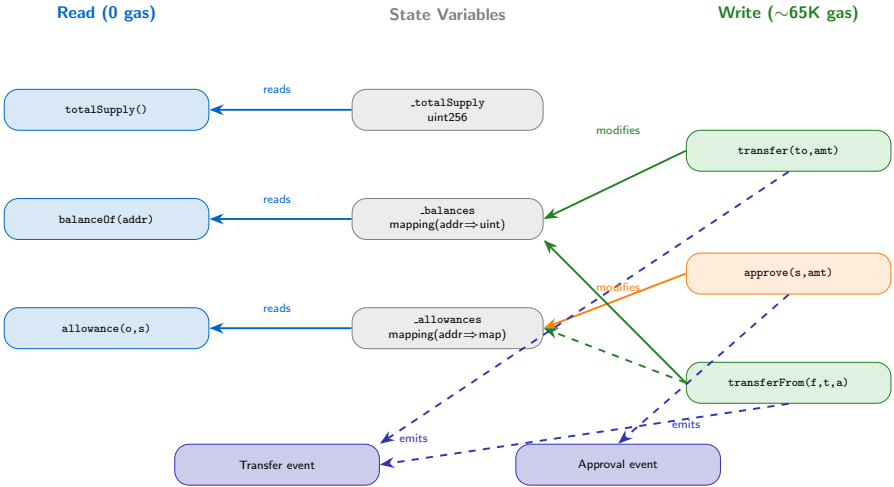
Address	Balance
0xAlice	499,900
0xBob	10,100
0xCarol	5,000

Total: 515,000 MTK

```
mapping(address => uint256) private _balances;
function transfer(address to, uint256 amount) public returns (bool) {
    require(_balances[msg.sender] >= amount, "Insufficient");
    _balances[msg.sender] -= amount;
    _balances[to] += amount;
    emit Transfer(msg.sender, to, amount);
    return true;
}
```

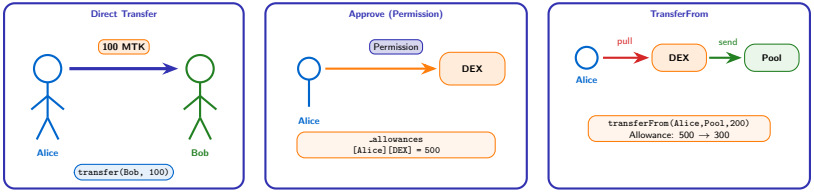
A token is just a mapping from addresses to balances — the smart contract is the ledger.

The ERC-20 Interface — All 6 Functions



Read functions cost zero gas; write functions require a transaction and pay gas.

Transfer vs Approve — The Full Mechanism



Race condition warning: set allowance to 0 first, then set to new value

```
mapping(address => mapping(address => uint256))
    private _allowances;
function approve(address spender, uint256 amount)
    public returns (bool) {
    _allowances[msg.sender][spender] = amount;
    emit Approval(msg.sender, spender, amount);
    return true;
}
```

approve()+transferFrom() is the delegation pattern that powers all of DeFi.

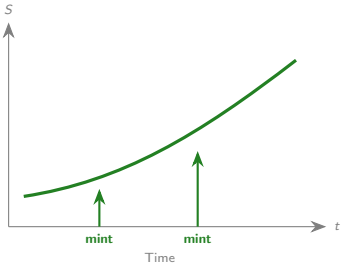
Token Supply Models — The Mathematics

Fixed Supply

$$S(t) = S_0, \quad \forall\, t \geq 0$$

```
uint256 public constant
  MAX_SUPPLY = 1_000_000 * 10**18;
```

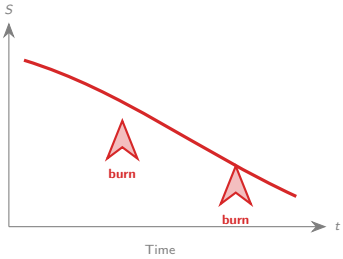
Inflationary



$$S_n = S_0 + \sum_{i=1}^{n} m_i$$

```
function mint(address to, uint256 a)
  onlyOwner { _mint(to, a); }
```

Deflationary

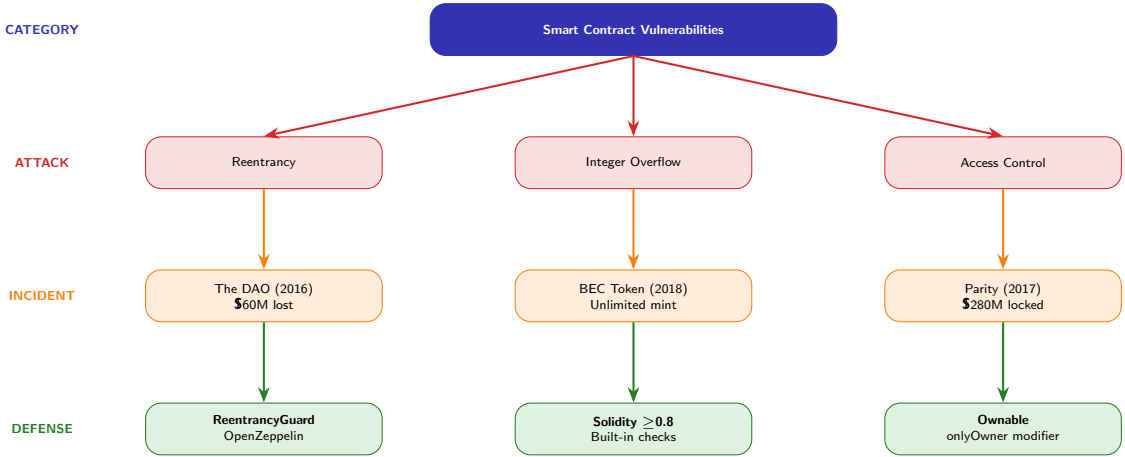


$$S_n = S_0 - \sum_{i=1}^{n} b_i$$

```
function burn(uint256 amt) public {
    _burn(msg.sender, amt); }
```

Tokenomics = mathematical models governing supply evolution via mint and burn events.

Security — Why OpenZeppelin?



Smart contract bugs are permanent and immutable — use audited libraries, always.

Key Terms for Beginners

Term	Definition
Blockchain	A shared, tamper-proof digital ledger that records every transaction across a network of computers.
Ethereum	A blockchain platform that supports smart contracts and is home to most ERC-20 tokens.
Smart Contract	A self-executing program on the blockchain that enforces rules automatically, without intermediaries.
Token	A digital asset created by a smart contract on an existing blockchain (unlike a coin, which has its own chain).
ERC-20	The standard interface (6 functions + 2 events) that all fungible tokens on Ethereum must implement.
Fungible	Interchangeable; one token equals any other of the same type, like dollar bills.
Solidity	The programming language used to write smart contracts on Ethereum.
Remix IDE	A free, browser-based tool for writing, compiling, and deploying Solidity smart contracts.
Gas	A fee (paid in ETH) for executing operations on Ethereum; every transaction costs gas.
Wallet	Software (e.g., MetaMask) that stores your private keys and lets you interact with the blockchain.

Term	Definition
Deploy	Sending your compiled contract to the blockchain, creating it at a unique address forever.
Testnet	A practice blockchain (e.g., Sepolia) where you can deploy and test for free, without real money.
DEX	Decentralized Exchange. A protocol (e.g., Uniswap) that lets users trade tokens directly from their wallets, without intermediaries.
OpenZeppelin	A library of audited, battle-tested smart contract code that prevents common security bugs.
Mapping	A Solidity data structure (like a dictionary) that stores key-value pairs, e.g., address $\rightarrow$ balance.
Mint	Creating new tokens, increasing the total supply. Typically restricted to the contract owner.
Burn	Permanently destroying tokens, decreasing the total supply.
Approve	Granting another address (e.g., a DEX) permission to spend your tokens on your behalf.
Tokenomics	The economic design of a token: supply model, distribution, incentives, and value.
Etherscan	A block explorer website where you can view transactions, contracts, and token balances.

Master these 20 terms and you have the vocabulary to understand any ERC-20 project.

Key Takeaways

Deploy Your Own Cryptocurrency: Summary

1. **Deploy in 15 minutes:** OpenZeppelin + Remix + Sepolia = your own token live on-chain.
2. **The balance mapping is the heart:** `mapping(address => uint256)` is the entire ledger.
3. **6 functions + 2 events = universal compatibility** with every wallet and DEX.
4. **approve()+transferFrom() enables DeFi composability** — the delegation pattern.
5. **Supply model math determines token economics** — choose fixed, inflationary, or deflationary.

Next: Token economics deep dive and real-world deployment strategies.

You now have the knowledge to create, deploy, and explain your own ERC-20 token.